

Rishabh Shah

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EDUCATION

Rutgers University – New Brunswick <i>M.S. in Computer Science, GPA: 3.83/4.0</i>	Expected May 2027
University of Mumbai <i>B.Tech. in Artificial Intelligence & Data Science, CGPA: 7.76/10.0</i>	Dec 2021 – Jun 2025 <i>Mumbai, India</i>

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL
Backend & APIs: Django, Django ORM, REST APIs, JWT, OAuth2, RBAC, Zod
Frontend: React, Next.js, Tailwind CSS, HTML, CSS
Databases & Data: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Prisma
Cloud & DevOps: AWS EC2, AWS S3, Google Cloud Run, Vercel, Docker, GitHub Actions, Git, Linux
Testing: Pytest, unittest, Vitest, Playwright, React Testing Library, Postman

EXPERIENCE

Aaditya Technologies <i>Software Engineering Intern</i>	Mumbai, India <i>Jan 2025 – Jul 2025</i>
- Developed service-oriented backend modules using Python, Next.js, and MySQL for authentication, authorization, business workflows, and reporting across internal and client-facing systems - Implemented JWT-based authentication and RBAC middleware for admin, staff, and client-user roles, securing protected API routes and centralizing authorization checks across backend workflows - Optimized dashboard, listing, search, and reporting APIs by indexing database columns, rewriting SQL queries, adding pagination, trimming response payloads, and batching recurring calls, reducing average response time by 22% - Containerized backend services with Docker and supported AWS EC2-hosted environments with S3-backed storage by configuring environment variables, debugging logs, and standardizing deployment steps, lowering setup effort by 65%	
Creative Line <i>Web Developer Intern</i>	Mumbai, India <i>Jun 2024 – Sep 2024</i>
- Built React/Next.js interfaces for dashboard views, data management pages, and form-based CRUD flows using reusable tables, forms, modals, filters, and pagination controls - Refactored page-level UI and request-handling logic into 7 shared components across 3 feature areas, shrinking individual feature file size by 70% and improving maintainability - Improved MySQL-backed Next.js API routes with parameterized queries, input validation, indexing, connection pooling, and smaller payloads, bringing P95 latency for targeted data-loading paths from 3100ms to 420ms	

TECHNICAL PROJECTS

CseGraph Local-First Code Context Engine & Developer Tooling	<i>May 2026</i>
- Engineered a local-first Python developer tool spanning a CLI, Model Context Protocol (MCP) stdio server, SDK facade, SQLite graph index, and VS Code wrapper to generate compact repository context for coding agents - Designed a Tree-sitter indexing pipeline for Python, JavaScript, and TypeScript that extracts files, symbols, imports, calls, inheritance, tests, and source references into a transactional SQLite graph - Delivered adaptive retrieval with FTS/BM25, graph proximity, token budgets, and strict schemas to return task-scoped context across 9 CLI commands and 6 MCP tools	
SnapInterview Backend-Heavy Full-Stack AI Interview Platform	<i>Feb 2026</i>
- Architected a full-stack Next.js and TypeScript platform using layered API routes, service logic, repository modules, Prisma models, and PostgreSQL persistence to support interview sessions, transcripts, feedback, users, and admin operations - Implemented secure account lifecycle features with Better Auth, Google OAuth, OTP email verification, password reset, account lockout, protected route gating, and admin-only access controls to protect user and admin actions - Built an end-to-end interview pipeline for resume PDF parsing, streaming LLM responses via Groq, transcript and evaluation persistence, shareable results, and S3 multipart recording uploads for reviewable mock interviews	
Traffic Violation Detection & Automated Ticketing System Computer Vision Backend Pipeline	<i>May 2025</i>
- Orchestrated a modular Python processing pipeline that coordinated red-light, helmet, and speeding detection workflows before routing violation data through OCR, validation, PDF generation, cloud storage, and SMS delivery - Integrated Llama 4 Scout via Groq for license plate OCR, regex-based license plate validation, vehicle registration verification, Jinja2/pdfkit document generation, AWS S3 object storage, and Twilio messaging into an automated ticketing workflow - Designed stateful video-processing logic using ByteTrack object tracking, per-vehicle state management, duplicate-ticket suppression, and frame-skipping to support automated speeding enforcement	

PUBLICATIONS & CERTIFICATIONS

Intelligent Traffic Surveillance – IEEE ICICIS 2025, Scopus indexed JARVIS: Voice Assistant with Smart Home Automation – ISTE 2025 AI-ML Virtual Internship – AICTE & EduSkills, supported by Google for Developers Generative AI Virtual Internship – AICTE & EduSkills, curriculum by Google Cloud	
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